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CS11021– Programming Laboratory (Intake 43)

Lab Sheet 5

Part I

- Identify required inputs by understanding the problem description
- Obtain **all necessary values from user input**
- Use the given **expected output format** to understand what needs to be displayed
- Follow the output format strictly for all questions
- Select appropriate data types for each input and calculation
- Use type casting where required to produce correct results
- Display outputs clearly using proper formatting

1. A cricket training center records player match data digitally. After each practice match, the system generates a performance report that includes a numerical efficiency value derived from the player's performance. The program should read the required information from the user and display a clear performance summary with calculated values displayed accurately.

```
Player Performance Report
-------------------------
Player Name :
Runs Scored :
Balls Faced :
Strike Rate :
```

2. An electronics showroom wants to display real-time power consumption details of its devices. When a device is connected, the system reads electrical measurements and presents the computed power value in a format suitable for very large or very small numbers.

```
Electronic Device Details
-------------------------
Device Name         :
Power Consumption   :
```

3. An online shopping platform automatically generates a purchase invoice when a customer places an order. The invoice includes product details, quantity purchased, and a total cost value. For internal processing, the system also stores the same value as a whole number.

```
Online Shopping Invoice
-----------------------
Product Name        :
Quantity Purchased  :
Total Cost          :
Stored Cost Value   :
```

4. A video game records player performance statistics after each session. One of the key indicators shown to the player is accuracy, which represents performance efficiency. The system displays this value both as a precise measurement and as a whole-number summary.
   - ***(accuracy = successActions / totalActions * 100)***

```
Game Performance Summary
------------------------
Total Actions       :
Successful Actions  :
Accuracy (%)        :
Accuracy (Rounded)  :
```

5. A mobile network provider tracks customer data usage at the system level. The usage is initially recorded in the smallest unit, but the user application displays this information in multiple higher-level units for better readability.

```
Internet Usage Report
---------------------
Data Used (KB) :
Data Used (MB) :
Data Used (GB) :
```

6. A research laboratory logs experimental values digitally. Certain calculated measurements may result in extremely small or extremely large numerical values. The system must present these results in a standardized scientific format to ensure clarity and precision.

```
Scientific Measurement Log
--------------------------
Computed Result :
```

Part II

1. A traffic **control unit** stores information about **road signals** using **numerical values** entered by the operator. The system records the **current traffic light status** and **pedestrian signal status** and evaluates their **combined condition numerically**. The program processes the entered values and produces a **summarized status output** that represents the **current traffic situation** using **operator-based evaluation**.
   - ***(combinedStatus = trafficLight * 10 + pedestrianSignal)***

2. A **smart home controller** manages several **electronic devices** using a **single control value**. Each device is represented internally using **individual bits** within that value. The user provides the **initial device state** and performs **operations** to **activate** or **deactivate** selected devices. The program processes the **input values** and displays the **updated device control state** after applying the **required operations**.

3. An organization records **employee salary details digitally**. The **initial salary** is provided by the user, and the system applies **incremental updates** to represent **salary growth** over successive updates. The program displays how the **salary value changes** during each update operation and shows the effect of **increment operations** on the **stored salary value**.

4. In a **computer networking environment**, numerical values are used to represent **IP addresses** and **subnet masks**. The user enters the required **address information** using the **keyboard**. The system applies **bitwise operations** on the entered values to compute and display the **resulting network address** used for **internal routing purposes**.

5. Study the flowchart carefully before writing the program.
   - Identify all required inputs from the keyboard: **levels completed**, **points per level**, and **bonus multiplier**.
   - Calculate the **total points** using the values entered by the user.
   - Compute **bonus points** and display them in **scientific notation**.
   - Display all results clearly, following the flowchart sequence:
     - Levels completed
     - Points per level
     - Total points
     - Bonus points
   - Use **appropriate data types** for calculations.
   - Apply **type casting** wherever necessary to ensure correct calculations.
   - Do **not use selection (if) or repetition statements (for, while)**.
   - Format the output neatly to make it readable.
   - ***(totalPoints = levels * pointsPerLevel * bonusMultiplier)***

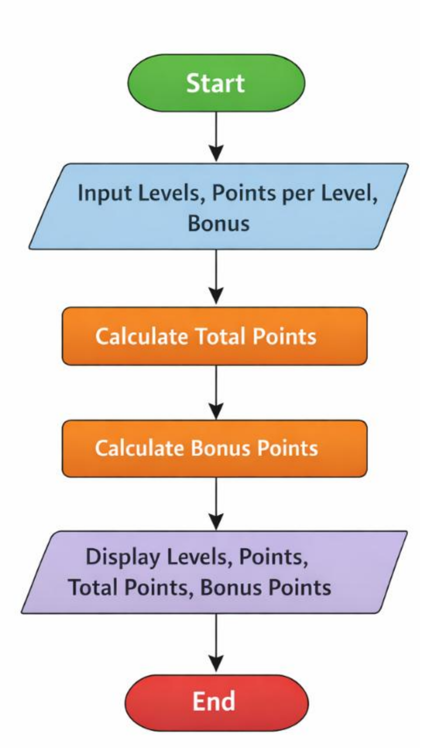

Figure 01: Game score calculation flowchart